Swagger
Supported by SMARTBEAR

/v3/api-docs

Explore

MetaboTrack API 1.0 OAS 3.0

/v3/api-docs

MetaboTrack API Documentation

Servers

http://localhost:8081 - Generated server url

Authorize

Record Management

Standardized endpoints for managing daily metabolic and activity data snapshots.

GET /api/record/{id} Get Record Details

Query detailed information based on the specified record ID.

Parameters

Try it out

Name	Description
id * required integer($int64) (path)	Unique identifier of the metabolic record *Example* : 1001 1001

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {
    "id": 0,
    "userId": 0,
    "recordDate": "2026-04-14",
    "stepsPerDay": 0,
    "activeMinutes": 0,
    "caloriesBurned": 0,
    "sleepHours": 0,
    "hydrationLiters": 0,
    "heartRateResting": 0,
    "heartRateAvg": 0,
    "continuousExerciseDays": 0,
    "efficiencyScore": 0,
    "calorieEfficiency": "Low Efficiency",
    "createTime": "2026-04-14T03:32:01.389Z",
    "updateTime": "2026-04-14T03:32:01.389Z"
  },
  "timestamp": 0
}
```

400	Bad Request Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

404	Not Found Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

Code	Description	Links
500	Internal Server Error Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

PUT `/api/record/{id}` Update Anomalous Record Data

Correct previously entered dynamic data (e.g., step count errors due to wearable sync delays).

Parameters Try it out

Name	Description
id * required `integer($int64)` *(path)*	ID of the record to be modified *Example*: 1001 1001

Request body required application/json

Example Value | Schema

```
{
  "stepsPerDay": 13000,
  "activeMinutes": 50,
  "caloriesBurned": 700,
  "sleepHours": 8,
  "hydrationLiters": 3,
  "heartRateResting": 60,
  "heartRateAvg": 110,
  "continuousExerciseDays": 6,
  "efficiencyScore": 9.2,
  "calorieEfficiency": "HIGH"
}
```

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": true,
  "timestamp": 0
}
```

400	Bad Request Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

404	Not Found Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

500	Internal Server Error Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
```

Code	Description	Links
	`{ "msg": "string", "data": {}, "timestamp": 0 }`	

GET `/api/record` Get Paginated Records

Retrieve a paginated list of metabolic records using DTO parameters.

Parameters Try it out

Name	Description
pageNum `string` *(query)*	Current page number *Example* : 1 1
pageSize `string` *(query)*	Number of records per page *Example* : 10 10
userId `string` *(query)*	Filter by Login in User ID userId
calorieEfficiency `string` *(query)*	Filter by specific calorie efficiency rating (1=LOW, 2=MODERATE, 3=HIGH) calorieEfficiency

Responses

Code	Description	Links
200	OK Media type	*No links*

Code	Description	Links

/

Controls Accept header.

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {
    "total": 0,
    "records": [
      {}
    ]
  },
  "timestamp": 0
}
```

400 Bad Request — *No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

404 Not Found — *No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

500 Internal Server Error — *No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

POST /api/record Create Daily Metabolic Record

Receives and saves a daily metabolic record of the user's metabolic metrics (e.g., exercise, heart rate, sleep).

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "userId": 1001,
  "recordDate": "2026-04-07",
  "stepsPerDay": 12500,
  "activeMinutes": 45,
  "caloriesBurned": 650.5,
  "sleepHours": 7.5,
  "hydrationLiters": 2.5,
  "heartRateResting": 62.5,
  "heartRateAvg": 115,
  "continuousExerciseDays": 5,
  "efficiencyScore": 8.75,
  "calorieEfficiency": "HIGH"
}
```

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": true,
  "timestamp": 0
}
```

Code	Description	Links
400	Bad Request Media type */* **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	*No links*
404	Not Found Media type */* **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	*No links*
500	Internal Server Error Media type */* **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	*No links*

DELETE `/api/record` Batch Deletion of Records

Hard delete multiple extreme anomalous data records simultaneously. Expects a JSON array of record IDs in the request body.

Parameters Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
[
  1001,
  1002,
  1005
]
```

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema `{ "code": 0, "msg": "string", "data": true, "timestamp": 0 }`	*No links*
400	Bad Request Media type */* **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	*No links*
404	Not Found Media type */*	*No links*

Code	Description	Links
	Example Value | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	

500	Internal Server Error
	
	Media type
	`*/*`
	
	Example Value | Schema
	
	`{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`
		No links

Advanced Analytics (Insights)

High-level data aggregation and predictive analytics endpoints based on population snapshots.

POST `/api/analytics/simulator` Dynamic Target Predictor & Simulator

A rule-based engine that accepts hypothetical daily targets (e.g., planned sleep, planned steps) and predicts the user's future efficiency score. Used for frontend interactive sliders and goal setting.

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "userId": 1001,
  "targetSteps": 10000,
  "targetActiveMinutes": 45,
  "targetCaloriesBurned": 2500,
  "targetSleepHours": 7.5,
  "targetHydration": 2.5,
```

```
  "currentStreak": 5,
  "restingHeartRate": 65,
  "averageHeartRate": 115
}
```

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {
    "baselineScore": 7.2,
    "projectedScore": 8.5,
    "scoreChange": "+1.3",
    "projectedTier": "HIGH",
    "feedbackMessages": "['Excellent sleep bonus applied.', 'Warning:
Low active minutes.']"
  },
  "timestamp": 0
}
```

400	Bad Request Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

404	Not Found Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
```

Code	Description	Links

```
  "data": {},
  "timestamp": 0
}
```

500	Internal Server Error	*No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

GET `/api/analytics/users/{userId}/percentile` Individual Percentile Ranking

Calculates where a specific user stands compared to the entire population. It aggregates the user's historical efficiency score and computes their percentile rank against all other users in the database, returning a gamified title.

Parameters Try it out

Name	Description
userId * required `integer($int64)` *(path)*	*Example* : 1001 1001

Responses

Code	Description	Links
200	OK	*No links*

Media type

/

Controls `Accept` header.

Code	Description	Links

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {
    "userId": 1001,
    "personalAvgScore": 8.95,
    "percentile": 85.5,
    "rankTitle": "ELITE (Top 15%)"
  },
  "timestamp": 0
}
```

400 Bad Request *No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

404 Not Found *No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

500 Internal Server Error *No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

GET `/api/analytics/population/efficiency-distribution`

Get Population Efficiency Distribution

Aggregates the 15,000-user snapshot dataset to calculate the distribution of efficiency tiers. Returns user counts and average physiological metrics per tier (LOW, MODERATE, HIGH) to support dashboard visualizations.

Parameters

Try it out

No parameters

Responses

Code	Description	Links
200	OK	*No links*

Media type

/

Controls Accept header.

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": [
    {
      "tier": "HIGH",
      "userCount": 4520,
      "avgSteps": 12450,
      "avgActiveMinutes": 55.5,
      "avgSleepHours": 7.8,
      "avgRestingHeartRate": 62.3
    }
  ],
  "timestamp": 0
}
```

400	Bad Request	*No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
```

Code	Description	Links

```
  "timestamp": 0
}
```

404	Not Found	*No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

500	Internal Server Error	*No links*

Media type

/

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

GET `/api/analytics/leverage/sleep-quality` Analyze Sleep Quality Leverage

Groups the population into sleep duration buckets (<6h, 6-8h, >8h) and calculates average exercise volume vs. average heart rate. Used to medically prove that insufficient sleep increases cardiovascular burden for the same amount of exercise.

Parameters | **Try it out**

No parameters

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": [
    {
      "sleepCategory": "NORMAL (6-8h)",
      "userCount": 8500,
      "avgSteps": 10500,
      "avgRestingHeartRate": 68.5,
      "avgDailyHeartRate": 115.2
    }
  ],
  "timestamp": 0
}
```

400	Bad Request Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

404	Not Found Media type */* **Example Value** | Schema	*No links*

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

500	Internal Server Error Media type */*	*No links*

Code	Description	Links

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": {},
  "timestamp": 0
}
```

GET /api/analytics/leverage/exercise-streak Analyze Exercise Streak Benefits

Groups the dataset by continuous exercise days (0, 1-3, 4-7, >7) to demonstrate how habits affect resting heart rate and metabolic efficiency scores. High-value insight for behavioral health analysis.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	OK	*No links*

Media type

/

Controls Accept header.

Example Value | Schema

```
{
  "code": 0,
  "msg": "string",
  "data": [
    {
      "streakCategory": "CONSISTENT (4-7 Days)",
      "recordCount": 3200,
      "avgRestingHeartRate": 64.2,
      "avgEfficiencyScore": 8.9,
      "avgCaloriesBurned": 580
    }
  ],
  "timestamp": 0
}
```

Code	Description	Links

400	Bad Request
	
	Media type
	/
	
	Example Value | Schema
	
	{
	"code": 0,
	"msg": "string",
	"data": {},
	"timestamp": 0
	}
		No links

404	Not Found
	
	Media type
	/
	
	Example Value | Schema
	
	{
	"code": 0,
	"msg": "string",
	"data": {},
	"timestamp": 0
	}
		No links

500	Internal Server Error
	
	Media type
	/
	
	Example Value | Schema
	
	{
	"code": 0,
	"msg": "string",
	"data": {},
	"timestamp": 0
	}
		No links

User Authentication

Endpoints for managing user accounts, including registration, login, and logout.

POST `/api/user/register` User Registration

Register a new user account. Passwords will be securely hashed before storage.

Parameters

Try it out

No parameters

Request body required

application/json

Example Value | Schema

```
{
  "username": "user_0",
  "password": "12345678"
}
```

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. Example Value | Schema `{ "code": 0, "msg": "string", "data": true, "timestamp": 0 }`	*No links*
400	Bad Request Media type */* Example Value | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	*No links*
404	Not Found Media type	*No links*

Code	Description	Links
	/ **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	
500	Internal Server Error Media type */* **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`	*No links*

POST `/api/user/logout` User Logout

Acknowledge logout request. The client application MUST discard the JWT locally to complete the logout process.

Parameters Try it out

No parameters

Responses

Code	Description	Links
200	OK Media type */* Controls `Accept` header.	*No links*

Code	Description	Links
	Example Value | Schema `{ "code": 0, "msg": "string", "data": "string", "timestamp": 0 }`	

400	Bad Request
	
	Media type
	`*/*`
	
	Example Value | Schema
	
	`{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`
		No links

404	Not Found
	
	Media type
	`*/*`
	
	Example Value | Schema
	
	`{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`
		No links

500	Internal Server Error
	
	Media type
	`*/*`
	
	Example Value | Schema
	
	`{ "code": 0, "msg": "string", "data": {}, "timestamp": 0 }`
		No links

POST /api/user/login User Login

Authenticate credentials and receive a JWT token for subsequent API requests.

Parameters Try it out

No parameters

Request body required application/json

Example Value | Schema

```
{
  "username": "user_1001",
  "password": "12345678"
}
```

Responses

Code	Description	Links
200	OK Media type */* Controls Accept header. **Example Value** | Schema `{ "code": 0, "msg": "string", "data": "string", "timestamp": 0 }`	*No links*
400	Bad Request Media type */* **Example Value** | Schema `{ "code": 0, "msg": "string", "data": {},`	*No links*

Code	Description	Links
	"timestamp": 0 }	

404	Not Found
	
	Media type
	/
	
	Example Value | Schema
	
	{
	"code": 0,
	"msg": "string",
	"data": {},
	"timestamp": 0
	}
		No links

500	Internal Server Error
	
	Media type
	/
	
	Example Value | Schema
	
	{
	"code": 0,
	"msg": "string",
	"data": {},
	"timestamp": 0
	}
		No links

Schemas